| Test Method for Colorfastness to Perspiration | Test method #: VF- E16<br>REF #: TM-R- 008<br>Page 1 of 3 |
|---|---|
| Date Issued: November 21, 2018<br>Revision #: 4<br>Reason for revisions: 1. Specify the requirement to always check the pH of the prepared solution > 2. Must calibrate the pH meter with the use of buffer solution. >> 3. Should remove the 8.0 pounds weight from the top of the perspiration tester during testing. | Created by: Nila Babia Erazo |

**PURPOSE AND SCOPE:** To measure the tendency of dyestuffs and other colorants to migrate from a test material, as a result of contact with artificial perspiration.

**REFERENCE:** AATCC 15

**EQUIPMENT AND MATERIAL:**

1. Multi-fiber fabric A
2. Immersion dish
3. Distilled water
4. Paper towel
5. Oven capable of maintaining a temperature of 38+/-2$^{o}$C.
6. AATCC Chromatic Transference Scale.
7. Perspiration tester B.
8. Sodium chloride (NaCl).
9. Lactic Acid, USP 85%
10. Disodium hydrogen phosphate, anhydrous (Na2HPO4)
11. Histidine monohydrochloride.
12. Sodium Hydroxide (NaOH)
13. Buffer solutions preferably pH (4, 7, 10, 12)
14. pH meter
15. Fabric, Leather, and synthetic leather

**SAMPLE PREPARATION:**

***Condition the sample for at least 24 hours at standard laboratory environment at 23.0±2.0˚C temperature and 55 ±10 % Relative Humidity.***

A. Test Sample Lay-up:

Die or hand cut a piece of the test material and a multi-fiber fabric a minimum of 2 cm in width by 5 cm in length, to a maximum of 5.7cm x 5.7cm. Place the test material directly on top of the multi-fiber fabric, (If both sides of the test material are to be evaluated, test the second side separately from the first. That is, prepare a second test material /multi-fiber lay-up with the other side of the test material contacting the multi-fiber fabric, and test separately.)

| Test Method for Colorfastness to Perspiration | Test method #: VF- E16<br>REF #: TM-R- 008<br>Page 2 of 3 |
|---|---|
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The following surfaces should be tested:

**Fabric =** Top side / finished surface only

**Leather (linings)** = Grain side only

**Synthetic (linings)** = Both sides

**B. • Artificial Perspiration Solution for pH 4.3 ±0.2**

Prepare one liter of artificial perspiration as follows:

10 grams of sodium chloride (NaCl)

1 gram of lactic acid, USP 85%

1 gram of disodium hydrogen phosphate, anhydrous (Na2HPO4)

0.25gram histidine monohydrochloride

Add distilled water as necessary to obtain 1.0 liter of solution

**Notes:**

- ***Ensure that the prepared solution has the correct pH. Adjust the pH by adding NaOH to increase the pH or lactic acid to lower the pH.***
- ***pH meter should be calibrated with the use of buffer solution***
- The prepared artificial perspiration solution should be discarded **not later than 3 days.**

**REPETITIONS:** Test one (1) sample per material.

| Test Method for Colorfastness to Perspiration | Test method #: VF- E16<br>REF #: TM-R- 008<br>Page 3 of 3 |
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**TEST PROCEDURE:**

1. Immerse the test material and multi-fiber fabric lay-up in the artificial perspiration test solution at 23+/-2 ° C. Place a glass or plastic plate directly on top of the lay-up to ensure that the test material and multi-fiber fabric remain in contact and are fully sub-merged in the water. Agitate the artificial perspiration solution occasionally to insure thorough wetting. Remove the test sample lay-up from the water after 15 minutes.
2. Remove excess artificial perspiration solution from the test sample lay-up by lightly wiping the surface with a paper towel or tissue. Whenever possible, the wet weight of the test sample should be 2.5-3 times its dry weight.
3. Immediately place the test material and multi-fiber fabric lay-up between two of the plastic plates from the perspiration tester. Insert the plastic plates into the perspiration tester, and place the metal compensating plate and the 8.0 pounds (3.6kg) weight on top of the plastic plates and achieve total of download weight at 4.5 kg. (Note: the weight of compensating plate is approx. 0.9 kg or 2.0 pounds.)

   ***Note**: Multiple test materials can be tested at the same time by stacking them on top of each other in the perspiration tester, provided a plastic plate always separates each of the test material lay-ups.*
4. Place the perspiration tester in an oven at 38+/-2 ° C for 6 hours. ***Ensure to remove the 8.0 pounds weight from the top of the perspiration tester.***
5. Separate the test material from the multi-fiber fabric and completely dry at a temperature no greater than 60 ° C.
6. Evaluate colorfastness by rating the color of each of the multi-fiber test fabrics in comparison to the AATCC Chromatic Transference Scale. The evaluation must be performed under normal daylight.

***Note**: Chromatic Transference Scales must be replaced after 12 months from purchase.*

**Report**: The value of the lowest rated multi-fiber fabric to the nearest 0.5, according to the AATCC Chromatic Transference Scale. **Discoloration rating 1-5 AATCC scale.**